

T-POT HONEYPOT

DEPLOYMENT ON MICROSOFT AZURE

Complete Step-by-Step Technical Documentation

Platform	Microsoft Azure (Free Tier)
OS	Ubuntu Linux
Honeypot	T-Pot (All-In-One Multi Honeypot)
Source	github.com/telekom-security/tpotce
SSH Port	tcp/64295 (changed by installer)
Web UI Port	tcp/64297

For Educational & Research Purposes Only

Table of Contents

1. Project Overview	3
2. Prerequisites	3
3. Phase 1 – Azure Account Setup	4
4. Phase 2 – Virtual Machine Creation	5
5. Phase 3 – Network Security Configuration	8
6. Phase 4 – Connecting to the VM via SSH	9
7. Phase 5 – System Update	10
8. Phase 6 – T-Pot Installation	11
9. Phase 7 – Accessing the T-Pot Web Interface	15
10. Architecture Overview	16
11. Troubleshooting	17

1. Project Overview

This documentation covers the complete end-to-end deployment of T-Pot — the All-In-One Multi Honeypot Platform — hosted on a Microsoft Azure Virtual Machine running Ubuntu Linux. The project demonstrates cloud infrastructure setup, network security configuration, and cybersecurity threat intelligence collection.

What is T-Pot?

T-Pot is an open-source honeypot platform developed by Deutsche Telekom Security. It bundles over 20 different honeypots in a single Docker-based deployment, combined with a powerful ELK stack (Elasticsearch, Logstash, Kibana) for visualization and analysis of attack data.

What is a Honeypot?

A honeypot is a security mechanism designed to detect, deflect, and study unauthorized access attempts. It mimics a legitimate target to attract attackers, capturing their tactics and techniques without exposing real systems.

Key Benefits of this Project

- Real-time visibility into global attack patterns and threat actors
- Hands-on experience with Azure cloud infrastructure
- Practical exposure to Docker containerization and Linux administration
- Free-tier Azure deployment (using \$200 credit) — no cost to start
- Rich Kibana dashboards for attack data visualization

2. Prerequisites

Before beginning, ensure you have the following:

Requirement	Details
Microsoft Account	Required to create an Azure free account
Credit/Debit Card	Needed for Azure free account verification (not charged)
Windows Terminal / PuTTY	SSH client to connect to the Azure VM
Internet Connection	Stable broadband recommended
Basic Linux Knowledge	Familiarity with terminal commands

3. Phase 1 – Azure Account Setup

STEP 1

Create a Free Azure Account

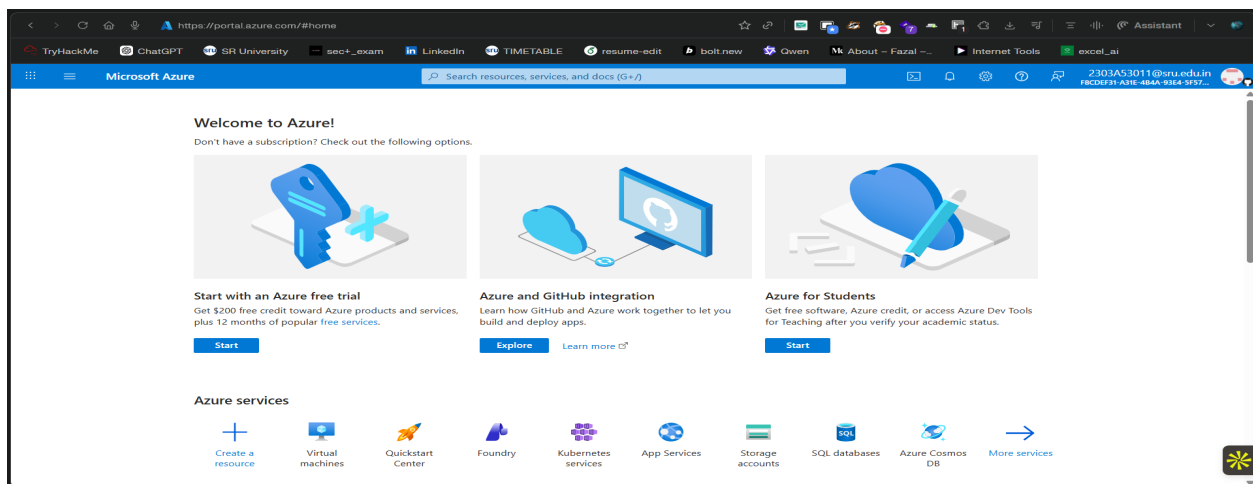
Microsoft Azure offers a free account with \$200 USD in credits valid for 30 days, plus always-free services. This is sufficient to run the T-Pot honeypot for the project duration.

1. Navigate to <https://azure.microsoft.com/free> in your web browser.
2. Click Start free and sign in with your Microsoft account (or create one).
3. Complete the identity verification — enter your phone number for SMS verification.
4. Enter your credit/debit card details for verification. You will NOT be charged unless you upgrade to a paid plan.
5. Agree to the terms and click Sign up.
6. Once the account is created, you will receive \$200 in credits and access to the Azure Portal at <https://portal.azure.com>.

Note

The \$200 credit is valid for 30 days from account creation.

The VM used in this project (Standard B2s or similar) will comfortably run within the free credit allocation.



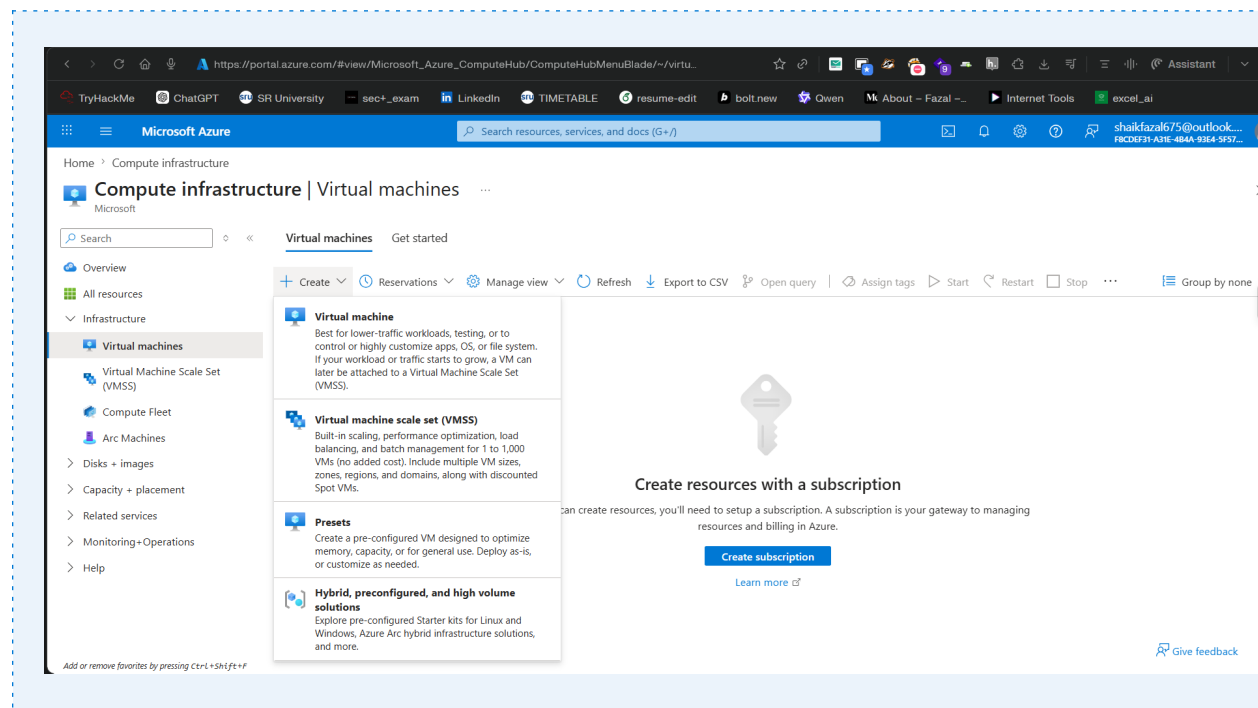
4. Phase 2 – Virtual Machine Creation

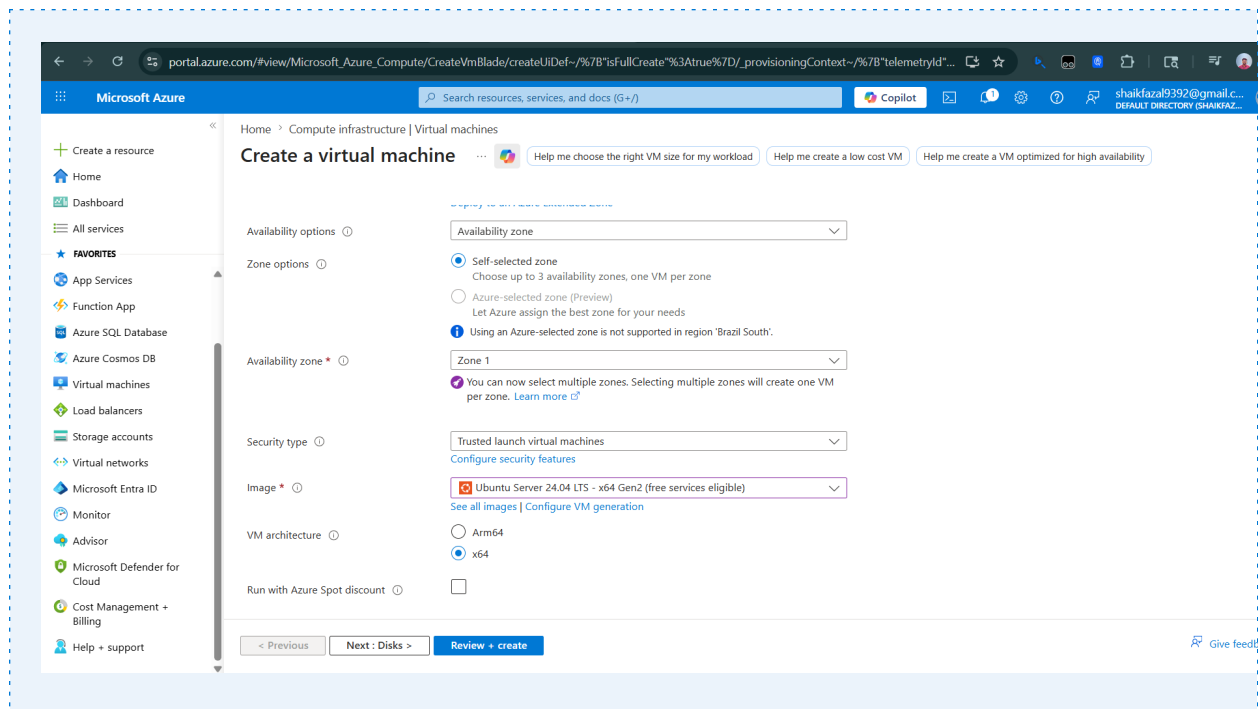
STEP 2

Create a New Resource Group

A Resource Group is a container that holds related Azure resources. All components of this project (VM, disk, network interface, public IP, etc.) will be placed in this group for easy management.

7. In the Azure Portal, click Create a resource in the top-left.
8. Search for Virtual Machine and select it.
9. Click Create → Azure virtual machine.
10. Under Resource group, click Create new and enter a name (e.g., HoneypotRG). Click OK.
11. Enter a VM name (e.g., tpot-vm).
12. Select your preferred Region (choose one geographically close for lower latency).





STEP 3

Configure VM Settings

Configure the operating system, size, and authentication settings for the Virtual Machine.

Image (Operating System)

- Image: Ubuntu Server 22.04 LTS - x64 Gen2
- Architecture: x64

VM Size

- Recommended: Standard_B2s (2 vCPUs, 4 GB RAM) or higher
- T-Pot requires at least 8 GB RAM for optimal performance — Standard_B2ms (2 vCPUs, 8 GB RAM) is preferred

Administrator Account (Authentication)

- Authentication type: Password
- Username: Choose a secure username (e.g., azureuser)
- Password: Set a strong password and note it down safely

Security Note

Use a strong password with a mix of uppercase, lowercase, numbers, and symbols.
This password will be used to SSH into the VM and also set as the T-Pot web UI credential.

machines for Linux images. [Learn more](#)

Administrator account

Authentication type ⓘ

☐ SSH public key

☒ Password

Username * ⓘ ✓

Password * ✓

Confirm password * ✓

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ

☐ None

☒ Allow selected ports

Select inbound ports *

STEP 4

Configure OS Disk Size

T-Pot stores considerable log data from honeypot activity. A larger OS disk is recommended to prevent storage issues during operation.

13. Navigate to the Disks tab in the VM creation wizard.
14. Under OS disk size, click Change size.
15. Select 256 GiB (or larger) from the dropdown.
16. Leave the OS disk type as Premium SSD or Standard SSD.
17. Leave all other disk settings at their defaults.

 Encryption at host is not registered for the selected subscription. [Learn more](#)

OS disk

OS disk size ⓘ 256 GiB (P15) ▼

 Some images are, by default, smaller than the selected OS disk size. [Click here to learn how to expand your disk partition size after you create your VM.](#)

OS disk type * ⓘ Premium SSD (locally-redundant storage) ▼

Delete with VM ⓘ ☒

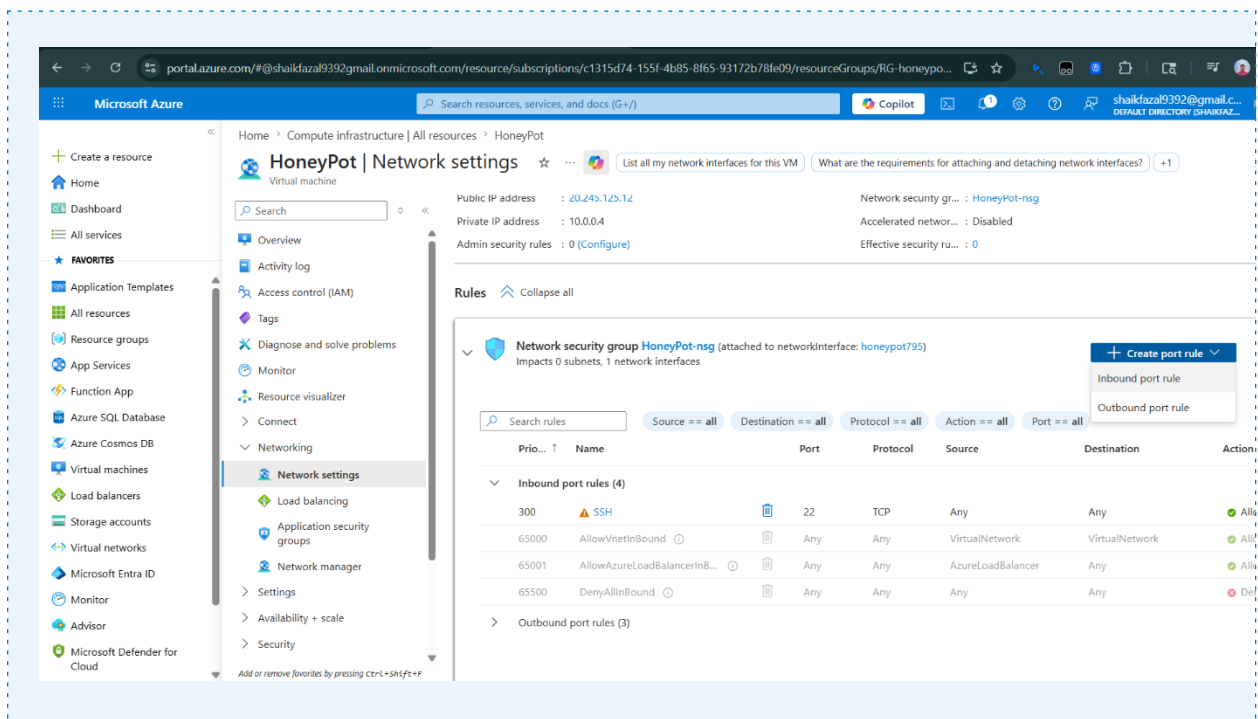
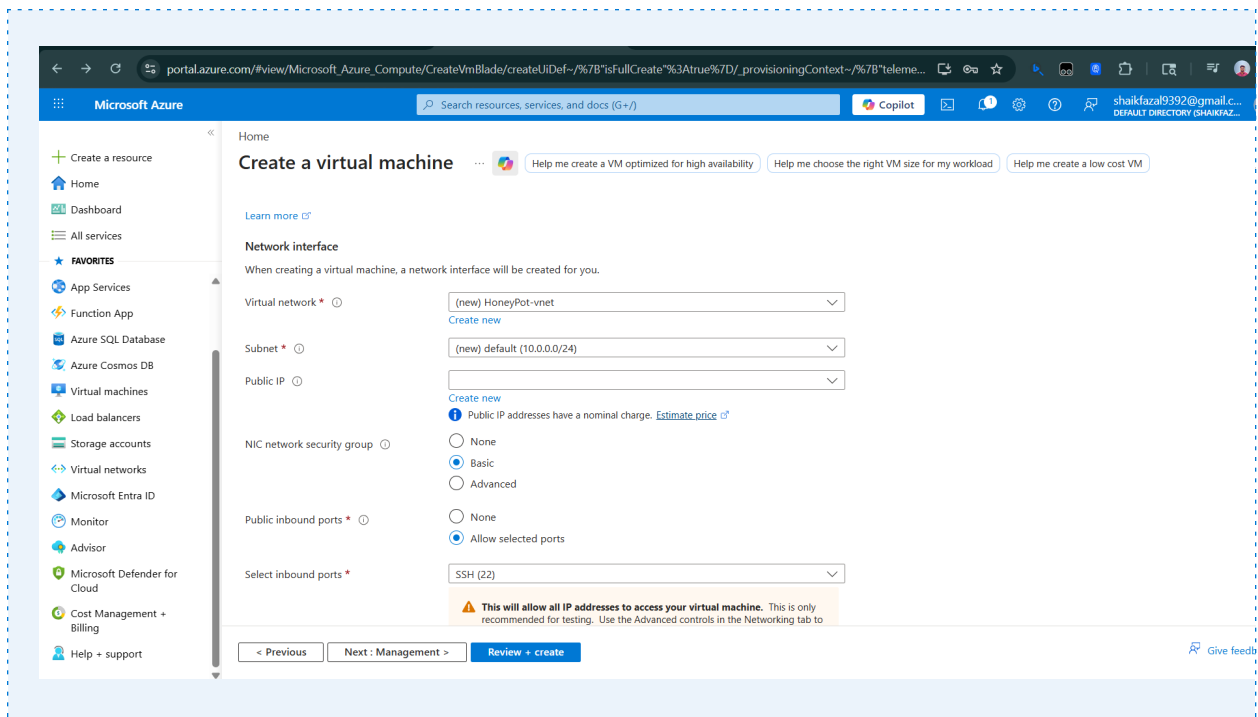
Key management ⓘ Platform-managed key ▼

Enable Ultra Disk compatibility ⓘ ☐

STEP 5

Review and Create the VM

18. Leave all other tabs (Networking, Management, Monitoring, etc.) at their default settings.
19. Click the Review + create tab.
20. Azure will validate the configuration. Wait for the "Validation passed" message.
21. Review the summary and estimated cost.
22. Click Create to begin the deployment.
23. Deployment takes approximately 2–5 minutes. You will see a "Your deployment is complete" message when done.
24. Click Go to resource to navigate to your new VM.



5. Phase 3 – Network Security Configuration

STEP 6

Configure Inbound Network Rules

T-Pot uses a wide range of ports to simulate various services that honeypots listen on. To capture attacks from all directions, we configure Azure's Network Security Group (NSG) to allow all inbound traffic on all ports.

Important Security Warning

Opening all ports is intentional for a honeypot deployment — it maximizes attack surface for research. NEVER do this on a production system. This VM contains no real data or services. The VM is isolated within its own Resource Group for containment.

Steps to Add the Inbound Rule

25. From the VM overview page, click Networking in the left sidebar (under Settings).
26. Click on the Network security group name (e.g., tpot-vm-nsg).
27. In the NSG page, click Inbound security rules in the left menu.
28. Click + Add to create a new rule.
29. Configure the rule with these settings:

Setting	Value
Source	Any
Source port ranges	* (all)
Destination	Any
Service	Custom
Destination port ranges	* (all 65535 ports)
Protocol	Any
Action	Allow
Priority	100
Name	Allow-All-Inbound

30. Click Add to save the rule.



Add inbound security rule



HoneyPot-nsg

Source ⓘ

Any



Source port ranges * ⓘ

*

Destination ⓘ

Any



Service ⓘ

Custom



Destination port ranges * ⓘ

1-65535



Protocol



Any



TCP



UDP



ICMPv4



ICMPv6

Action



Allow



6. Phase 4 – Connecting to the VM via SSH

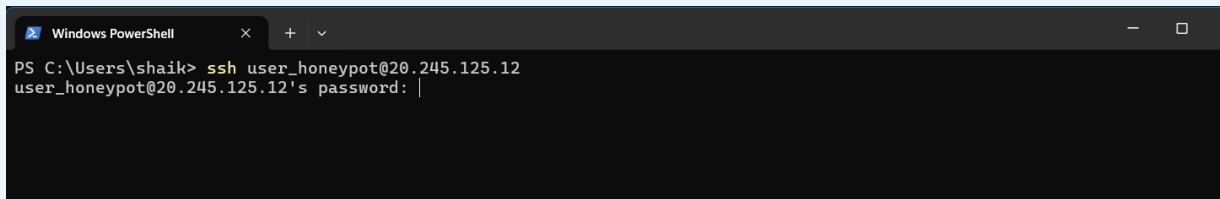
STEP 7

Connect to Azure VM Using Windows Terminal

With the VM running and the network rule configured, we can now SSH into the VM using its public IP address.

Finding the Public IP Address

31. Navigate to your VM in the Azure Portal.
32. On the VM Overview page, locate the Public IP address field on the right side.
33. Copy the IP address (e.g., 20.x.x.x).



```
Windows PowerShell
PS C:\Users\shaik> ssh user_honeypot@20.245.125.12
user_honeypot@20.245.125.12's password: |
```

Connecting via SSH

34. Open Windows Terminal (or Command Prompt / PowerShell).
35. Type the following SSH command, replacing the values with your own:

```
$ ssh <your-username>@<public-ip>

# Example:
$ ssh azureuser@20.123.45.67
```

36. When prompted "Are you sure you want to continue connecting?", type yes and press Enter.
37. Enter your VM password when prompted.
38. You are now connected to your Azure Ubuntu VM.

7. Phase 5 – System Update

STEP 8

Update Ubuntu to the Latest Version

Before installing T-Pot, it is critical to update all system packages to ensure compatibility and security. Run the following commands in your SSH session:

```
# Update package list
$ sudo apt update

# Upgrade all installed packages to latest versions
$ sudo apt upgrade -y

# Optional: Remove unnecessary packages
$ sudo apt autoremove -y
```

Wait for the update process to complete. This may take 3–10 minutes depending on the number of packages and network speed.

Tip

If prompted during upgrade about configuration files (e.g., `/etc/ssh/sshd_config`), select 'keep the local version currently installed' unless you have specific requirements.

8. Phase 6 – T-Pot Installation

STEP 9

Clone the T-Pot Repository

T-Pot is distributed via GitHub. Clone the official repository to the VM:

```
# Clone the T-Pot repository
$ git clone https://github.com/telekom-security/tpotce

# Navigate into the cloned directory
$ cd tpotce
```

A screenshot of a terminal window with a dark background. The window has tabs at the top labeled 'Vaults', 'SFTP', and '20.245.125.12'. The terminal shows the command 'user_honeypot@HoneyPot:~\$ git clone https://github.com/telekom-security/tpotce.git' being entered. The command is highlighted in green.

```
user_honeypot@HoneyPot:~$ git clone https://github.com/telekom-security/tpotce.git
```

STEP 10

Run the T-Pot Installer

Execute the installer script as a non-root user (with sudo privileges):

```
$ ./install.sh
```

What the Installer Does Automatically

The install.sh script performs the following actions on your Ubuntu system:

- 1 Changes the SSH port from 22 to tcp/64295 (to avoid conflicts with honeypots)
- 2 Disables the DNS Stub Listener to prevent port conflicts
- 3 Sets the firewall target for the public zone to ACCEPT
- 4 Adds Docker's official repository and installs Docker Engine
- 5 Installs all recommended packages (grc, exa, micro, etc.)

6	Removes packages known to cause conflicts
7	Adds the current user to the docker group (allows Docker without sudo)
8	Creates dps and dpsw aliases for docker ps monitoring
9	Creates la, ll, ls aliases using exa (improved ls command)
10	Creates mi alias for micro text editor
11	Displays open ports on the host for conflict checking
12	Adds and enables tpot.service in systemd for auto-start on boot

STEP 11

Follow the Installer Prompts

During the installation, you will be prompted to make several choices:

39. Enter your sudo password when requested (at least once during the process).
40. Select the T-Pot edition when prompted. Options include:
 - Standard — Full installation with all honeypots and dashboards (recommended)
 - Sensor — Only honeypots, no web interface
 - Industrial — SCADA/ICS focused honeypots
 - Collector — Log shipping only
41. Create a T-Pot username and password for the web interface (Kibana / Attack Map).
42. Monitor the installation output for any errors. The process may take 10–20 minutes.


```
127.0.0.1 : ok=42 changed=24 unreachable=0 failed=0 skipped=1 rescue=0 ignored=1

### Playbook was successful.

### Choose your T-Pot type:
### (H)ive - T-Pot Standard / HIVE installation.
###         Includes also everything you need for a distributed setup with sensors.
### (S)ensor - T-Pot Sensor installation.
###         Optimized for a distributed installation, without WebUI, Elasticsearch and Kibana.
### (L)LM - T-Pot LLM installation.
###         Uses LLM based honeypots Beelzebub & Galah.
###         Requires Ollama (recommended) or ChatGPT subscription.
### M(i)ni - T-Pot Mini installation.
###         Run 30+ honeypots with just a couple of honeypot daemons.
### (M)obile - T-Pot Mobile installation.
###         Includes everything to run T-Pot Mobile (available separately).
### (T)arpit - T-Pot Tarpit installation.
###         Feed data endlessly to attackers, bots and scanners.
###         Also runs a Denial of Service Honeypot (ddospot).

### Install Type? (h/s/l/i/m/t) h

### Installing T-Pot Standard / HIVE.

### T-Pot User Configuration ...

### Enter your web user name: fazal_honeytrap
```

STEP 12

Reboot the Virtual Machine

After installation completes, reboot the VM to apply all changes and start T-Pot services:

```
$ sudo reboot
```

The SSH connection will drop immediately as the system restarts. Wait approximately 2–3 minutes for the VM to fully restart and all Docker containers to initialize.

After Reboot

SSH will now be available on port 64295 (not the default port 22).

To reconnect after the reboot:

```
$ ssh -p 64295 <your-username>@<public-ip>
```

Note: The Azure NSG all-ports-open rule allows this new port without any changes.

9. Phase 7 – Accessing the T-Pot Web Interface

STEP 13

Access the T-Pot Dashboard

Once the VM has rebooted and T-Pot services have started, you can access the web interface from any browser.

Opening the T-Pot Web UI

43. Open a web browser on your local machine.
44. Navigate to the following URL (replace with your VM's actual public IP):

```
https://<your-public-ip>:64297
```

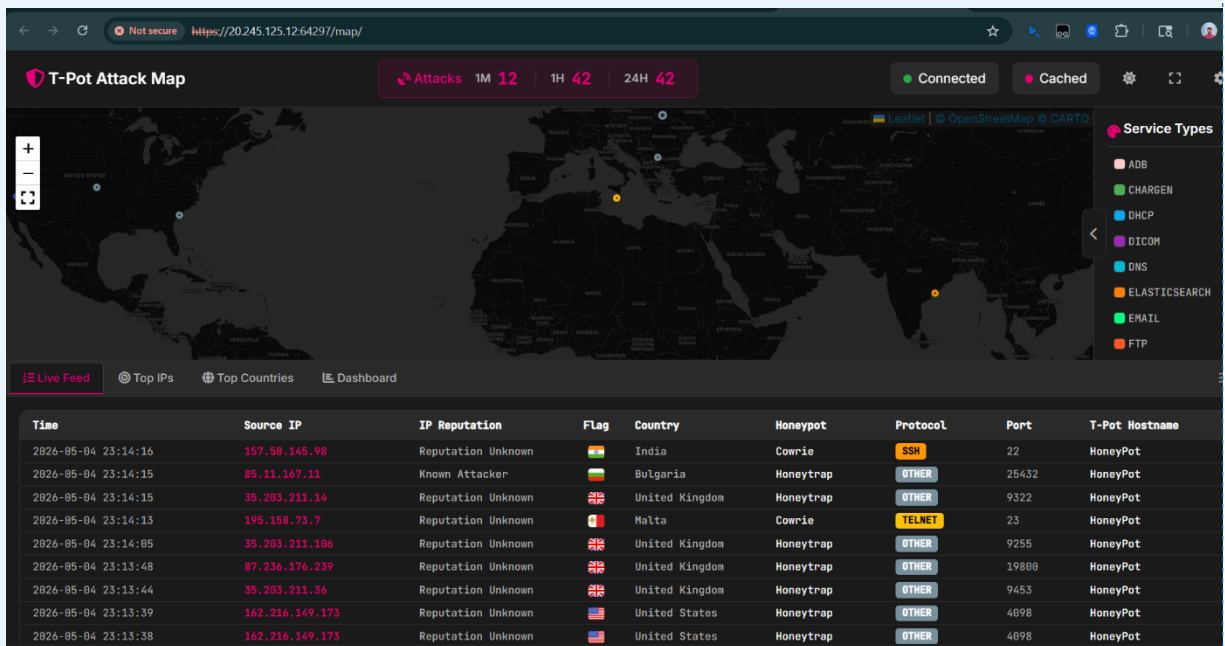
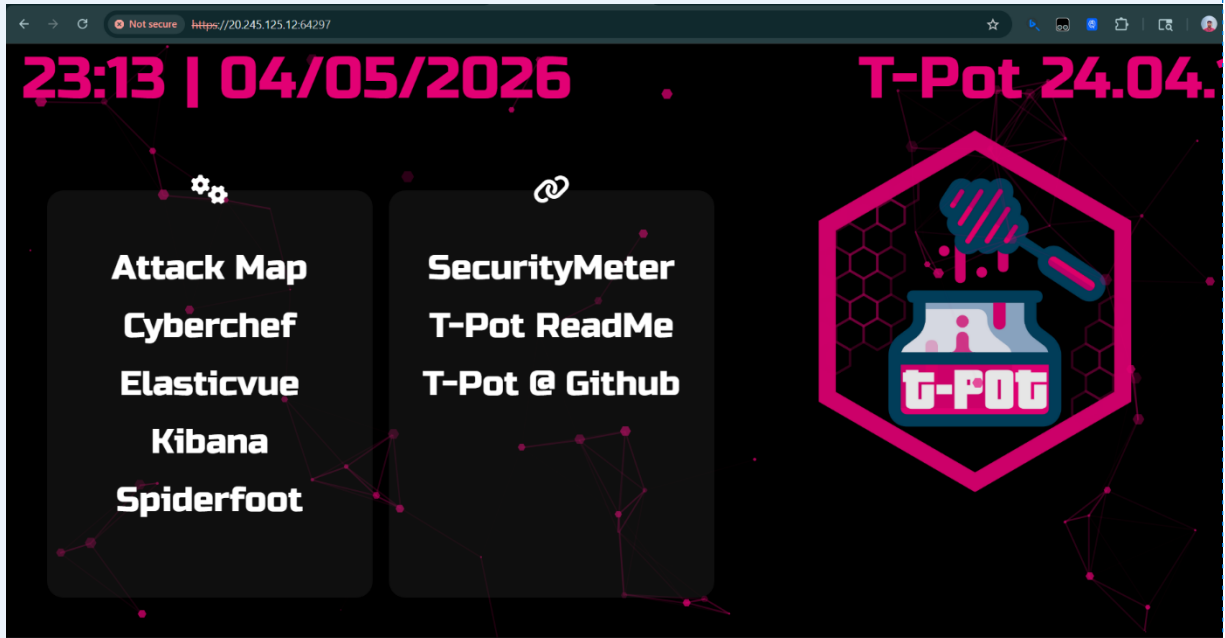
```
# Example:
```

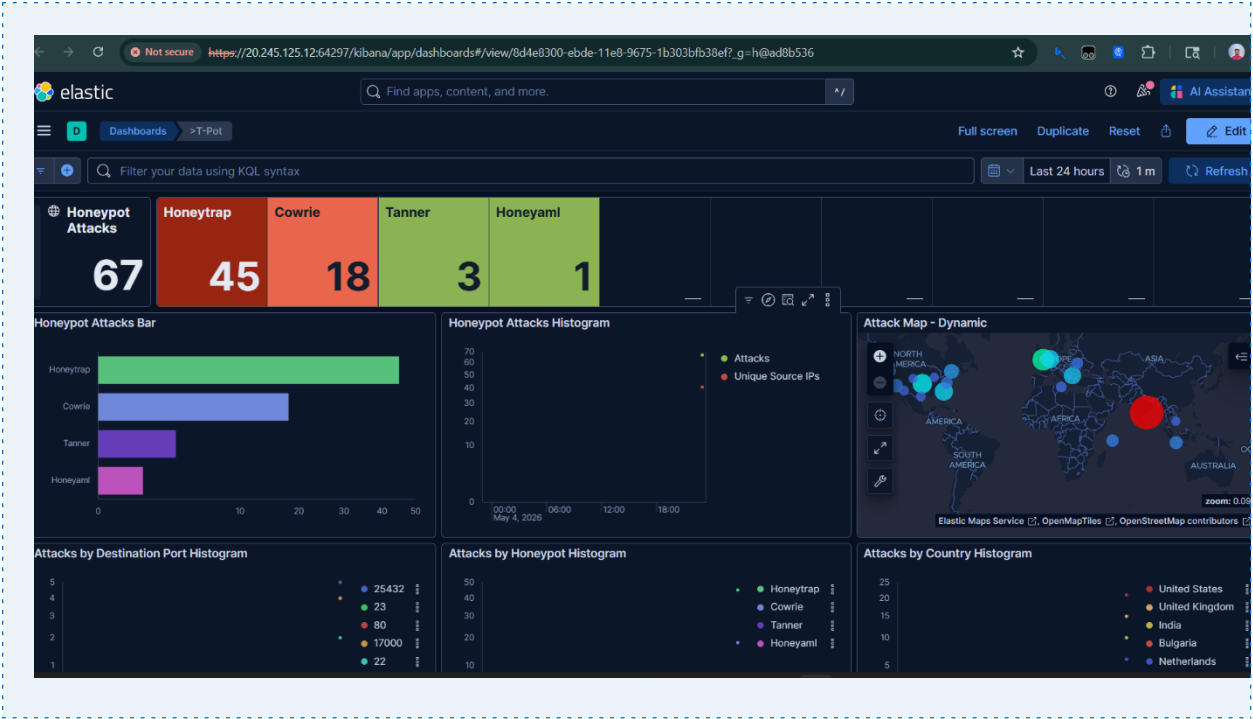
```
https://20.123.45.67:64297
```

45. The browser will show a security warning because T-Pot uses a self-signed SSL certificate. This is normal and safe in this context.
46. Click Advanced → Proceed to <IP> (or similar, depending on your browser) to continue.
47. A login dialog box will appear. Enter the T-Pot username and password you created during installation.
48. You will be presented with the T-Pot landing page with links to all available dashboards.

Available Dashboards & Tools

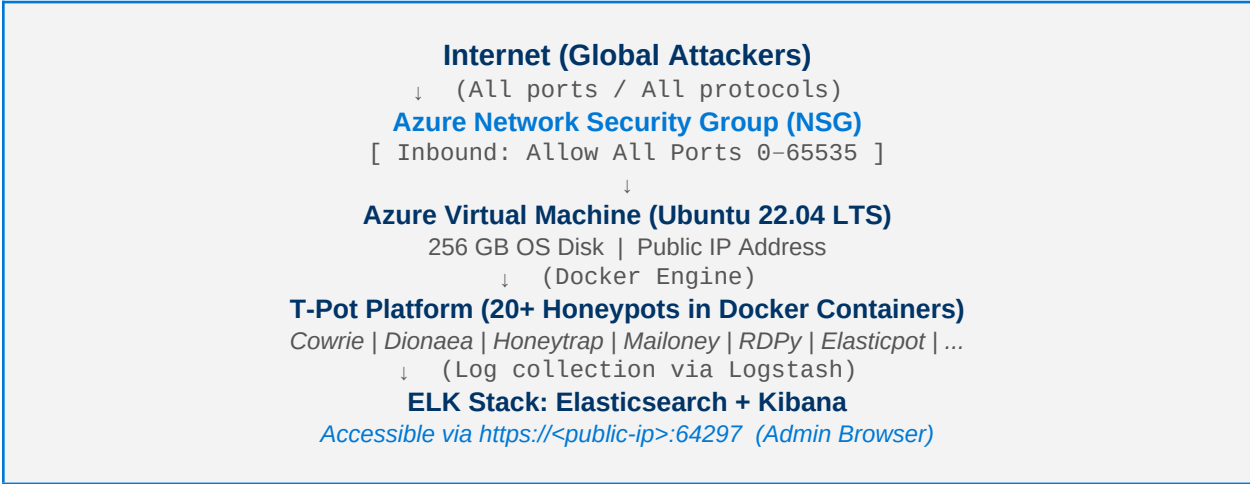
Tool	Description
Kibana	Primary dashboard for log analysis, attack patterns, and geolocation maps
Attack Map	Real-time live visualization of incoming attacks on a world map
Cyberchef	Data encoding/decoding and analysis tool
Elasticvue	Elasticsearch cluster management and data browser
Spiderfoot	OSINT and threat intelligence aggregation tool





10. Architecture Overview

The following diagram illustrates the complete architecture of the T-Pot honeypot deployment on Azure:



Key Ports Reference

Port	Protocol	Purpose
64295	TCP	SSH access to VM (changed from default 22)
64297	TCP (HTTPS)	T-Pot Web UI / Kibana Dashboard
0-65535	All	Honeypot listener ports (all open for attack capture)

11. Troubleshooting

Issue	Solution
Cannot SSH after reboot	Use port 64295: <code>ssh -p 64295 user@<ip></code> . T-Pot changes SSH from port 22.
Web UI not reachable	Wait 3–5 minutes after reboot. All Docker containers need time to start.
SSL certificate warning in browser	Normal for self-signed cert. Click Advanced → Proceed to continue.
Installation failed midway	Check disk space (<code>df -h</code>). Ensure at least 50 GB free. Retry <code>./install.sh</code> .
Docker containers not starting	Run: <code>sudo systemctl status tpot</code> to see error details and logs.
Cannot clone git repo	Ensure git is installed: <code>sudo apt install git -y</code>
Azure credits running low	Stop the VM when not actively monitoring to save credits.